

HAO HU

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SUMMARY

Senior backend engineer with deep CS fundamentals and architecture judgment, now an AI-focused CTO — 10+ years building large-scale distributed systems and data infrastructure, including at Google, Uber, and Microsoft.

Production AI / agentic systems builder — RAG, conversational analytics, and MCP-backed multi-agent workflows with observability, context controls, deterministic guardrails, and human approval for risky actions.

Hands-on engineering leader — leads small teams from design through delivery, using spec-driven, review-gated AI-assisted workflows with automated tests and evals to ship faster without sacrificing reliability.

TECHNICAL SKILLS

LLM & Agents: agent orchestration, agent harnesses, tool / function calling, MCP integration, multi-agent routing, context engineering / memory, LangChain / LangGraph / DeepAgents, OpenAI / Claude / DeepSeek

RAG & Retrieval: RAG (multimodal), hybrid retrieval, embeddings, reranking, query optimization, Milvus, Qdrant

Evaluation & Safety: LangSmith, RAGAS, eval harnesses, LLM-as-judge, guardrails, human-in-the-loop

Programming Languages: Java & Go (**Google readability**), Python, SQL, C++, C#

Backend & Distributed Systems: microservices, API design, gRPC / Protobuf, REST, Thrift, Spring AI, FastAPI, SSE

Data Systems: data modeling, CDC, PostgreSQL, MySQL, Redis, BigQuery, Kafka, Flink, ClickHouse, MongoDB

Cloud & Platform: Google Cloud, AWS, Docker, Kubernetes, Linux, Nginx, CI/CD

EXPERIENCE

Yuanbao Creative Technology

Chief Technology Officer

Feb 2025 – Present

Wuhan, China

- **Architected and shipped a unified AI layer for an ERP SaaS platform** across operations, analytics, and knowledge; established shared eval/safety practices: observability, golden sets, regression suites, and least-privilege tools.
- **Operations Copilot:** built a **DeepAgents**-based agent harness with delegated specialists (sales / inventory / order / purchasing), using role-scoped **MCP** tools, sandboxed execution, proactive monitoring, auditable tool traces, and a **Spring AI / Java 21** server with human-in-the-loop approval for risky writes (cryptographically bound, single-use).
- **Analytics Agent:** delivered self-service, multi-turn NL-to-SQL over the ERP **ClickHouse** warehouse, backed by a **Qdrant** semantic layer that cut prompt context ~73%; added OLAP intent routing, deterministic **SQLGlue** guardrails, read-only execution, bounded SQL repair, result-equivalence regression evals, and auto-charting.
- **Knowledge Assistant:** delivered **LangGraph**-orchestrated enterprise-document RAG with multi-source ingestion and **Milvus** hybrid retrieval; routed precise / balanced / broad strategies with query expansion and rerank, enforcing citation grounding, strict-evidence refusal, and RBAC; added LLM-judge/citation evals and per-query observability.

Independent

Independent Software Engineer & Consultant

Mar 2023 – Jan 2025

Wuhan, China

- **Built and deployed a cloud-native travel-guide SaaS** featuring user-uploaded audio/video POI guides with Google Maps-assisted GPS and trip planning, on AWS (EC2/Docker, RDS Postgres, S3, CloudWatch) and Vercel.
- **Built AI engineering depth** through RAG/agent prototypes and applied learning, bridging into the CTO role.

Uber

Senior Software Engineer

May 2022 – Mar 2023

Toronto, ON

- **Built Uber's first CPG (consumer-packaged-goods) Ads feature — storefront banner ads:** as backend lead on a 3-person team, built a Go/gRPC + DocStore service between UberEats Storefront and Ads so advertisers could run budgeted item/brand campaigns across retail chains; **scaled to ~250K QPS at ~140 ms p99** within a quarter.
- **Redesigned the CPG campaign model:** replaced restaurant-era single-location item IDs with a GTIN-keyed multi-location model — enabling national campaigns, removing millions of duplicates, and forming the CPG Ads foundation.
- **Launched store-menu sponsored items** by extending the CPG Ads backbone into a third-party ad integration: ranked sponsored items in storefront sub-menus, streamed impression/click events through a Kafka + Flink billing pipeline, and **held ~180 ms p99** with Prometheus / Grafana monitoring and dependency alerting.
- **Drove the staged production rollout of CPG Ads** and served as the Ads team's dev representative — aligning product, data science, sales, DevOps, partner teams, and the external ad provider.

Google

Feb 2018 – Feb 2021

Software Engineer — Cloud SQL (Data Integration / Disaster Recovery)

Sunnyvale, CA

- **Raised backup health from 99.992% to 99.997%** across a large fleet backing a 99.95% SLA: owned and hardened the backup-verification pipeline (snapshot attach, DB boot, checksum on pooled GCE instances), resolving top failure classes including resource exhaustion, startup failures, and false-positive verification failures.
- **Enabled PITR-safe transaction-log retention**: built backup PD-snapshot cleanup that pruned nonessential general-log data, preserved recovery-critical logs, and allowed restore to any timestamp via nearest backup + log replay.
- **Kept managed instances recoverable**: replaced a fixed-count retention window that pruned oldest-first regardless of health with a policy that always retained the most recent backup marked healthy by the verification pipeline.
- **Automated MySQL HA migration to Persistent Disk synchronous replication**: built a Java workflow converting legacy streaming master/replica pairs, spanning both control and data planes.
- **Built customizable CSV formatting across MySQL and PostgreSQL**: extended the backend REST API and redesigned import / export one-shot tasks in Go, enabling reliable export / import round-trips.
- **Shipped Cloud SQL PostgreSQL CSV import / export to GA**: hardened the injection-sensitive import / export path with on-demand, least-privilege database access and Cloud SQL IAM for Google Cloud Storage flows.
- **Supported Cloud SQL production reliability**: served on-call across the Java backend and Go on-instance agents, resolving release-blocking and customer-escalated issues, and mentored junior engineers.

Microsoft

Jan 2014 – Oct 2017

Software Development Engineer

Mountain View, CA

- **Co-built “People I Communicate With” (PICW) from the ground up** on a 3-person team: a foundational C# Outlook signal-collection component capturing frequent contacts and mail-interaction signals for Contact Safety and Exchange anti-spam, contributing to safety-level classification, junk-rule processing, and spam/phishing detection.
- **Took sole ownership of PICW and drove cross-team adoption of its signals**: served as point of contact for consuming teams, fed communication signals into People Relevance, powering people ranking for the Graph People API (people suggestions) and lightweight CRM scenarios.
- **Built a distributed C++ profiling tool that drove account selection for the Hotmail / Outlook.com-to-Exchange migration**: profiled feature usage across the 2.2B-account base, surfaced 40+ feature gaps, identified 400M+ migration-ready accounts, and re-profiled as gaps closed for successive migration waves.
- **Built monitoring for People Relevance**: real-time alerting, performance counters, and log-aggregation dashboards that became the team’s first-line signal for production health, improving triage and on-call response.
- **Drove Outlook service health**: served on-call across Hotmail / Outlook services, triaged release-blocking/customer issues, and shipped fixes to production.

Epic Systems

Jun 2012 – Nov 2013

Software Developer

Madison, WI

- **Built and extended EMC2**: Epic’s internal software configuration management platform (VB.NET on InterSystems Caché) for version control, dependency management, and automated build and deployment — an early CI/CD system.
- **Developed customer release and upgrade utilities** (console and web) that automated version checks and pushed updates over a secure communication framework, keeping health systems on current, supported versions.
- **Extended a Trial Dev Environment** for safe high-risk change integration and resolved build / deployment escalations.

HCR ManorCare

Jan 2011 – Apr 2012

Data Warehouse Developer

Toledo, OH

- **Built SSIS ETL pipelines feeding the enterprise data warehouse of a 500+ facility national healthcare provider**: extracted, validated, transformed, and loaded data from multiple sources into conformed fact/dimension schemas.
- **Developed SSRS reports and dashboards** (cross-tab, drill-down, parametric, cascaded) for finance/operations BI.
- **Created SQL Server objects** (stored procedures, indexed views, UDFs) powering report logic and ETL validation.

EDUCATION

Bowling Green State University

M.S. in Computer Science

Aug 2010 – May 2012

GPA 4.0/4.0 · Bowling Green, OH

Wuhan University

B.S. in Software Engineering

Aug 2006 – Jun 2010

GPA 3.5/4.0 · Wuhan, China